CASE STUDY 3

Hospital Billing and Patient Classification

Python Activity-Based Learning

1. Problem Statement

A hospital needs a simple program to calculate a patient's bill. The bill includes consultation, diagnostic tests, and treatment charges. The program applies a concession based on the patient's category and classifies the final payment amount according to a simple payment policy.

2. Objectives

• Calculate the total hospital bill from three types of charges.

• Apply a concession for eligible patients.

• Check that the entered charges are not negative.

• Classify the final bill as Low, Medium, or High.

• Display a clear bill summary.

3. Assumptions

• All charges are entered in rupees and must be zero or greater.

• Senior citizens receive a 10% concession on the total bill.

• Other patients do not receive a concession.

• After concession, a bill below ■5,000 is Low; ■5,000 to ■15,000 is Medium; above ■15,000 is High.

• This is an educational example, not a real hospital billing policy.

4. Algorithm

1. Start the program.

2. Input consultation, diagnostic, and treatment charges.

3. If any charge is negative, display an invalid-input message and stop.

4. Add all charges to get the total bill.

5. Ask whether the patient is a senior citizen.

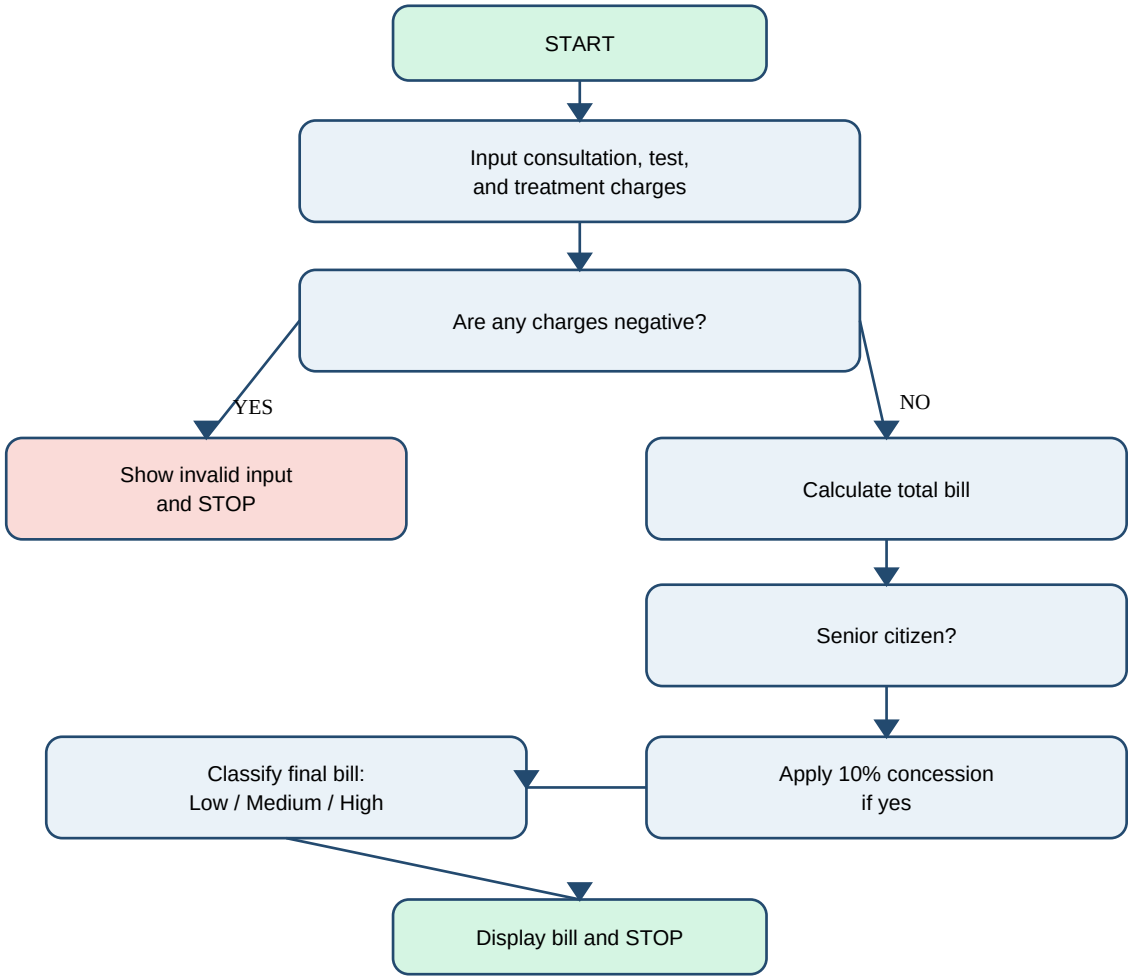
6. If yes, calculate a 10% concession; otherwise, concession is zero.

7. Subtract the concession from the total bill.

8. Classify the final bill as Low, Medium, or High.

9. Display the bill details and stop.

5. Flowchart



6. Pseudocode

```
START
INPUT consultation, tests, treatment
IF consultation < 0 OR tests < 0 OR treatment < 0
    PRINT "Invalid charges"
    STOP
total = consultation + tests + treatment
INPUT senior_citizen
IF senior_citizen == "yes"
    concession = total * 0.10
ELSE
    concession = 0
final_bill = total - concession
IF final_bill < 5000
    category = "Low"
ELSE IF final_bill <= 15000
    category = "Medium"
ELSE
    category = "High"
PRINT total, concession, final_bill, category
STOP
```

7. Python Source Code

```
consultation = float(input("Enter consultation charge: "))
tests = float(input("Enter diagnostic test charges: "))
treatment = float(input("Enter treatment charges: "))

if consultation < 0 or tests < 0 or treatment < 0:
    print("Invalid charges. Enter zero or positive values.")
else:
    total = consultation + tests + treatment
    senior = input("Is the patient a senior citizen? (yes/no): ").lower()
```

```
if senior == "yes":
    concession = total * 0.10
else:
    concession = 0

final_bill = total - concession

if final_bill < 5000:
    category = "Low"
elif final_bill <= 15000:
    category = "Medium"
else:
    category = "High"

print("\n--- HOSPITAL BILL ---")
print("Total bill: Rs.", round(total, 2))
print("Concession: Rs.", round(concession, 2))
print("Final bill: Rs.", round(final_bill, 2))
print("Patient bill category:", category)
```

8. Test Cases

Test	Consultation	Tests	Treatment	Senior?	Expected final bill	Category
1	500	1000	2000	no	3500	Low
2	1000	2000	7000	yes	9000	Medium
3	2000	5000	15000	no	22000	High
4	-100	500	1000	no	Invalid input	Not classified

Sample Output (Test Case 2)

```
Enter consultation charge: 1000
Enter diagnostic test charges: 2000
Enter treatment charges: 7000
Is the patient a senior citizen? (yes/no): yes

--- HOSPITAL BILL ---
Total bill: Rs. 10000.0
Concession: Rs. 1000.0
Final bill: Rs. 9000.0
Patient bill category: Medium
```

9. Output Screenshots

For submission, run the Python code in VS Code and take a screenshot of the terminal showing the sample output above. The sample output is included here as a text preview; replace or supplement it with your own actual VS Code screenshot if your faculty requires a screenshot.

10. Conclusion

This program calculates a hospital bill, gives a simple senior-citizen concession, validates charge inputs, and classifies the final bill. It uses basic Python concepts such as input(), variables, arithmetic operators, if/elif/else conditions, and print().